

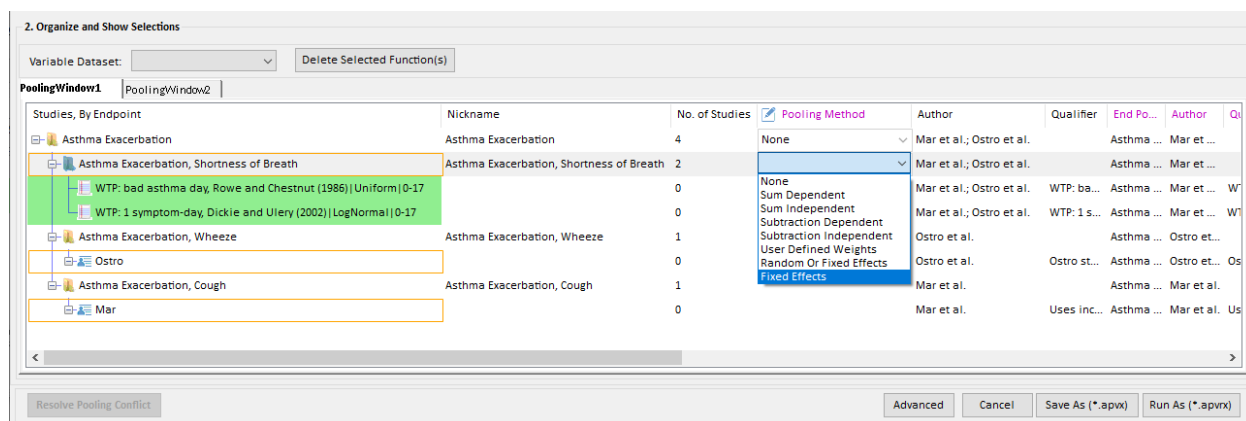
valuation points, then the holding container will have 1,000 values.) Finally, the holding container is sorted low to high and binned back down to 100 percentile points (representing the 0.5th percentile to the 99.5th percentile of the economic value of the incidence).

Step 2. Sort results

Depending on how your incidence results were pooled, the columns in the valuation pooling windows can be resorted in the same way as the **Select and Pool Incidence Estimates** window columns. This resorting will have the same sort of impact on the tree structure of valuation results that it had on the tree structure of incidence results. (See Step 2 in the section on Pooling Incidence Results.)

Step 3. Select pooling methods

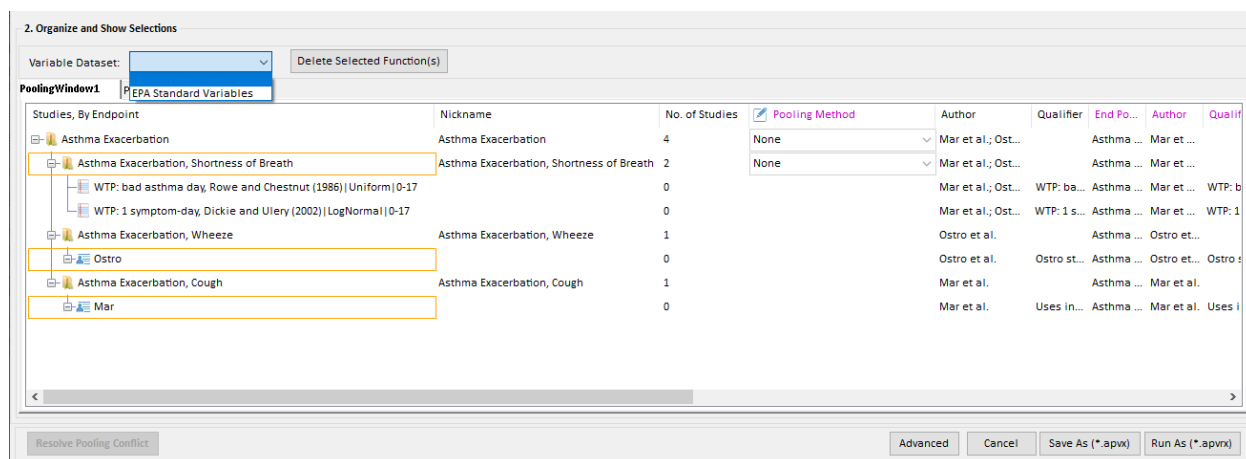
The same pooling methods are available for valuation results which were available for incidence results. (See Step 2 in the section on Pooling Incidence Results.) You should note that when more than one valuation method is selected for a particular pooled incidence result, it is possible to pool the generated valuation results.



Step 4. Select Variable Dataset

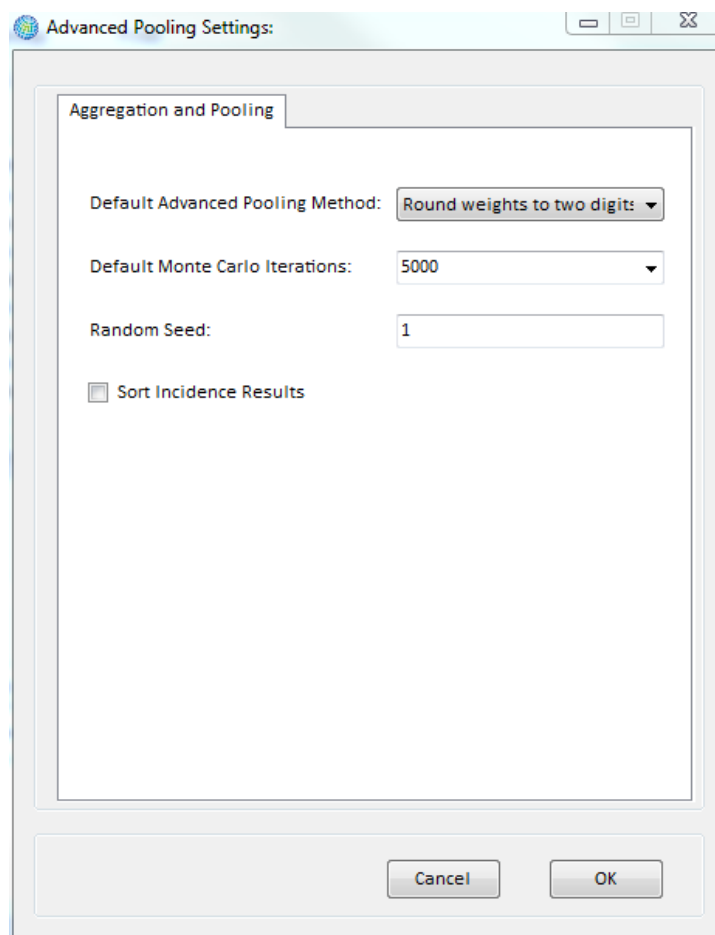
Before proceeding to the next step, you have the option to select a **Variable Dataset** from the drop-down menu beneath the pooling window (this is not required to proceed to the next step). The **Variable Dataset** can include a variety of data, such as income and poverty data that might be used in health or valuation functions. For the default EPA health and valuation functions, you just need to select the *EPA Standard Variables*.

If you have developed your own setup, then you have the option to load and use a **Variable Dataset**. This is not necessary if you do not need the extra variables that can be included in this dataset.



7.2.4 APV Configuration Advanced Settings

When specifying the incidence pooling options, you may click on the **Advanced** button on the bottom of the **Select and Pool Incidence Estimates** window. This button will open the **Advanced Pooling Settings** window. The features of this window are described in the next section.



7.2.4.1 Aggregation and Pooling

Default Advanced Pooling Method

The relative contribution of any one study in the pooling process depends on the weight assigned to that study. A key component of the pooling process, then, is the determination of the weight given to each study. BenMAP-CE lets users assign “subjective” weights and it assigns weights using a fixed effects or a random effects approach. There are three options for using weights available in the **Default Advanced Pooling Method** drop-down list:

Round weights to two digits. BenMAP-CE rounds each weight to two digits (e.g. 0.73), and then multiplies these weights by 100 to get two-digit integers. Each entire distribution (set of percentile points) is then put into a holding container an integral number of times, according to its integral weight. This holding container is then sorted low to high and binned down to the appropriate number of percentile points.

Round weights to three digits. BenMAP-CE rounds each weight to three digits (e.g. 0.732), and then multiplies these weights by 1,000 to get three-digit integers. Each distribution (set of percentile points) is then put into a holding container for an integral number of times, according to its integral weight. This holding container is then sorted low to high and binned down to the appropriate number of percentile points.

Use exact weights for Monte Carlo. BenMAP-CE uses exact weights and a Monte Carlo simulation. During each iteration of the procedure, a particular result is selected with a probability equal to its weight. Once a result is selected, one of its percentile points is chosen at random and put into a holding container. This is done some number of times (see *Monte Carlo Iterations*, below), and the holding container is then sorted low to high and binned down to the appropriate number of percentile points.

Default Monte Carlo Iterations

This drop-down list is only enabled when *Use exact weights for Monte Carlo* is selected as the **Default Advanced Pooling Method**. It specifies the number of iterations the Monte Carlo simulation should be run (see above). Its initial value is set by the **Default Monte Carlo Iterations** value from the **Advanced Pooling Settings** window (see Step 1, above).

Random Seed

The **Advanced Pooling Settings** window allows the specification of a **Random Seed**. Many of the pooling methods require the generation of sequences of random numbers, e.g. choosing a random percentile point during a Monte Carlo simulation. Providing a specific **Random Seed** value allows you to ensure that the same sequence of random

numbers is generated as in a previous analysis, thus allowing exact results to be reproduced.

If you do not set the **Random Seed** for a particular run, one will be generated automatically from the system clock (the number generated will depend on the date and time, and should change every minute). Normally, you should not set the **Random Seed** value. If you need to reproduce a specific set of results, however, the random seed used to generate previous APV Configuration Results can be determined from an APV Configuration Result file (.apvrx) Audit Trail Report.

Sort Incidence Results

The *Sort Incidence Results* should generally be always checked. This setting ensures that the incidence percentile-point results are sorted low to high.

7.2.4.2 Currency Year and Income Growth

The **Currency Year and Income Growth** window, accessed via the **Advanced** button on the **Valuation Method** window, allows you to specify an **Inflation Dataset** and a **Currency Year**, which in combination allow you to change the currency year to account for inflation. The **Income Growth Adjustment** panel allows you to adjust the valuation estimates to account for the growth in income over time for WTP functions.